

2.4 — The Overlap Move

Two figures that share a piece. One move finishes Example 7, Example 8, and half of 2.6. None of this is new math. It is learning to see the shared piece.

THE OVERLAP MOVE — find the shared piece first, every time

Given the WHOLEs are equal, prove the PARTS are equal:

break both wholes into pieces, then COVER the shared piece.

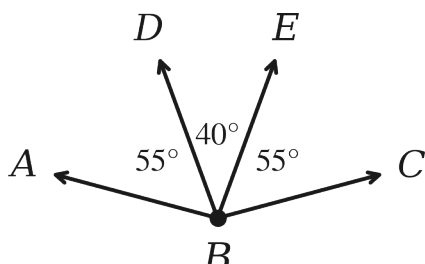
Given the PARTS are equal, prove the WHOLEs are equal:

ADD the shared piece to both, then name the two sums.

Either direction, the middle step is the same: say the given again, in pieces.

Part 1 — Check it with numbers first

Before proving anything, see that it is true.

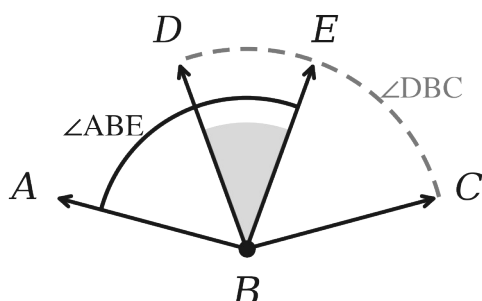


- $m\angle ABE = 55 + 40 = \underline{\hspace{2cm}}$
- $m\angle DBC = 40 + 55 = \underline{\hspace{2cm}}$
- So is $m\angle ABE = m\angle DBC$? $\underline{\hspace{2cm}}$
- And is $m\angle ABD = m\angle EBC$? $\underline{\hspace{2cm}}$
- The piece sitting inside BOTH big angles is $\angle \underline{\hspace{2cm}}$

Note: The two big angles are equal AND the two outer pieces are equal. That is not a coincidence. It happens because both big angles contain the same middle piece.

Part 2 — Name the pieces

The solid arc is $\angle ABE$. The dashed arc is $\angle DBC$. The shaded wedge sits inside both.



$\angle ABE$ is made of $\angle ABD$ and $\angle \underline{\hspace{2cm}}$

$\angle DBC$ is made of $\angle DBE$ and $\angle \underline{\hspace{2cm}}$

The reason that lets you break a whole angle into pieces: $\underline{\hspace{4cm}}$

$$m\angle ABD + m\angle DBE = m\angle DBE + m\angle \underline{\hspace{1cm}}$$

Note: Nothing new was deduced here. $m\angle ABE$ was only ever a NAME for $m\angle ABD + m\angle DBE$, and $m\angle DBC$ was only ever a name for $m\angle DBE + m\angle EBC$. You swapped both names for what they are made of. Same sentence, said in pieces.

 $m\angle$ _____

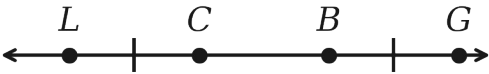
Part 5 — Now write the whole proof

Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.

Angle Addition Postulate — a whole angle is the sum of its pieces.
Segment Addition Postulate — a whole segment is the sum of its pieces.
Substitution Property of Equality — replace a name with what it equals.
Addition Property of Equality — put the same amount onto both sides.
Subtraction Property of Equality — take the same amount off both sides.

Part 6 — The same move, on segments

A library, a café, a bank and a gym sit in that order along a straight road. The library-to-café distance equals the bank-to-gym distance.



Given: $LC = BG$ Prove: $LB = CG$

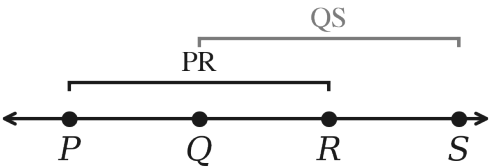
Note: Careful — this one runs the OTHER way. In Part 5 you were given that the two wholes were equal, and you covered the shared piece. Here you are given that the two small pieces are equal and you want the wholes, so you ADD the shared piece instead of covering it.

Which piece do LB and CG share? _____

Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.

Part 7 — On your own

No reason bank this time. Points P, Q, R and S lie in that order on a line.



Given: $PR = QS$ Prove: $PQ = RS$

Which piece do PR and QS share? _____

Cover the shared piece, or add it? _____

Statements	Reasons
1.	1.
2.	2.
3.	3.
4.	4.